

Exercise Sheet 2

System Safety

This sheet introduces key safety concepts and applies System-Theoretic Process Analysis (STPA) to the CARLA autonomous driving system described in the Course Roadmap. Refer back to that system description for all exercises below.

Hint: You are encouraged to solve these exercises digitally (e.g., using spreadsheets, digital diagrams, or notebooks). This will make it easier to later incorporate your results, diagrams, and evidence into your final safety report.

Terminology

Exercise 2.1: Safety Vocabulary

optional

Define the following terms in your own words and give a concrete example from an autonomous driving context for each:

1. **Loss:** An adverse consequence. Example: Injury or loss of human life.
2. **Hazard:** A system state or condition that can lead to a loss under certain circumstances. Example: A vehicle moving while a pedestrian is in its path.
3. **Risk:** The combination of the probability of a hazard, the probability of a loss given that hazard, and the severity. Formula:

$$\text{Risk} \approx P(\text{hazard}) \times P(\text{loss} \mid \text{hazard}) \times \text{Severity}$$

"This exercise establishes a common language for safety so we can distinguish between a bad outcome (Loss), the situation that makes it possible (Hazard), and the mathematical likelihood of it happening (Risk)."

STPA: System-Theoretic Process Analysis

STPA Step 1: Define Purpose and Identify Losses, Hazards, and ODD

Exercise 2.2: ODD Specification

Based *only* on the system description in the course Roadmap (you have not yet seen the training data), define the **Operational Design Domain** (ODD) for the CARLA system. Cover at least the following dimensions: weather, lighting, camera condition, scene type, and vehicle speed.

For each dimension, state:

1. The **operating conditions** (where the system is considered valid).
2. The **non-operating conditions** (where it is not).
3. How an ODD violation could be **detected at runtime**.

Hint: You are defining the ODD *before* looking at the data. In Exercise Sheet 3, you will compare this ODD to the actual training data and (possibly) discover gaps.

Dimension	Operating Condition (Valid)	Non-Operating Condition	Runtime Detection Method
Weather	Clear, cloudy (no rain/fog).	Heavy rain, fog, snow.	Anomaly/OOD detection scores.
Lighting	Daytime (high sun angle).	Night, low sun (glare).	Average pixel intensity.
Camera	Forward-facing, rigidly fixed.	Loose mount, lens obscuration.	Edge detection/Explainability.
Scene Type	Urban roads, mapped intersections.	Off-road, unmapped highways.	Scene classifier/OOD scores.
Vehicle Speed	≤ 50 km/h.	> 50 km/h.	Vehicle speedometer feedback.

"We define the Operational Design Domain to draw a clear boundary around where our model is 'allowed' to work, ensuring we don't expect it to handle conditions like heavy snow or night that it never saw during training."

Exercise 2.3: Losses

STPA begins by identifying the **losses** that must be prevented - unacceptable outcomes involving harm to people, property, or the environment.

For each loss, identify its ID (L-1, L-2, ...), briefly describe it, and explain why it is unacceptable:

"Safety engineering always starts at the end: we identify the absolute worst-case outcomes—like death or property damage—so we can build the entire system specifically to prevent them."

ID	Loss Description	Why Unacceptable
L-1	Injury or loss of human life.	Primary ethical and moral imperative of safety-critical systems.
L-2	Material damage (AV or property).	High financial repair costs and liability for the developer.
L-3	Violation of traffic laws/fines.	Leads to legal liability and potential loss of testing license.
L-4	Loss of public trust/Reputation.	Social and ethical backlash can end the entire AV program.

Exercise 2.4: Hazards

A **hazard** is a system state that, combined with worst-case environmental conditions, can lead to a loss. Hazards are properties of the *system*, not of individual components.

1. Identify at least four system-level hazards (H-1, H-2, ...) for the CARLA system.
2. For each hazard, state which loss(es) from Exercise 2.3 it can lead to.
3. Rate each hazard's likelihood (low / medium / high) and its severity (low / medium / high).

Fill in the following table:

ID	Hazard	Losses	Likelihood	Severity
H-1	Pedestrian in path while vehicle is moving.	L-1, L-2.	Medium	High
H-2	Autonomous mode remains active outside the ODD.	L-1, L-2, L-3.	High	Medium
H-3	Approaching intersection without detecting stop condition.	L-1, L-2, L-3	Medium	High
H-4	Planner commands acceleration when obstacle is in path.	L-1, L-2.	Low	High

"Hazards bridge the gap between normal driving and a crash; they identify the specific system states, like moving while a pedestrian is in the way, that lead directly to our identified losses."

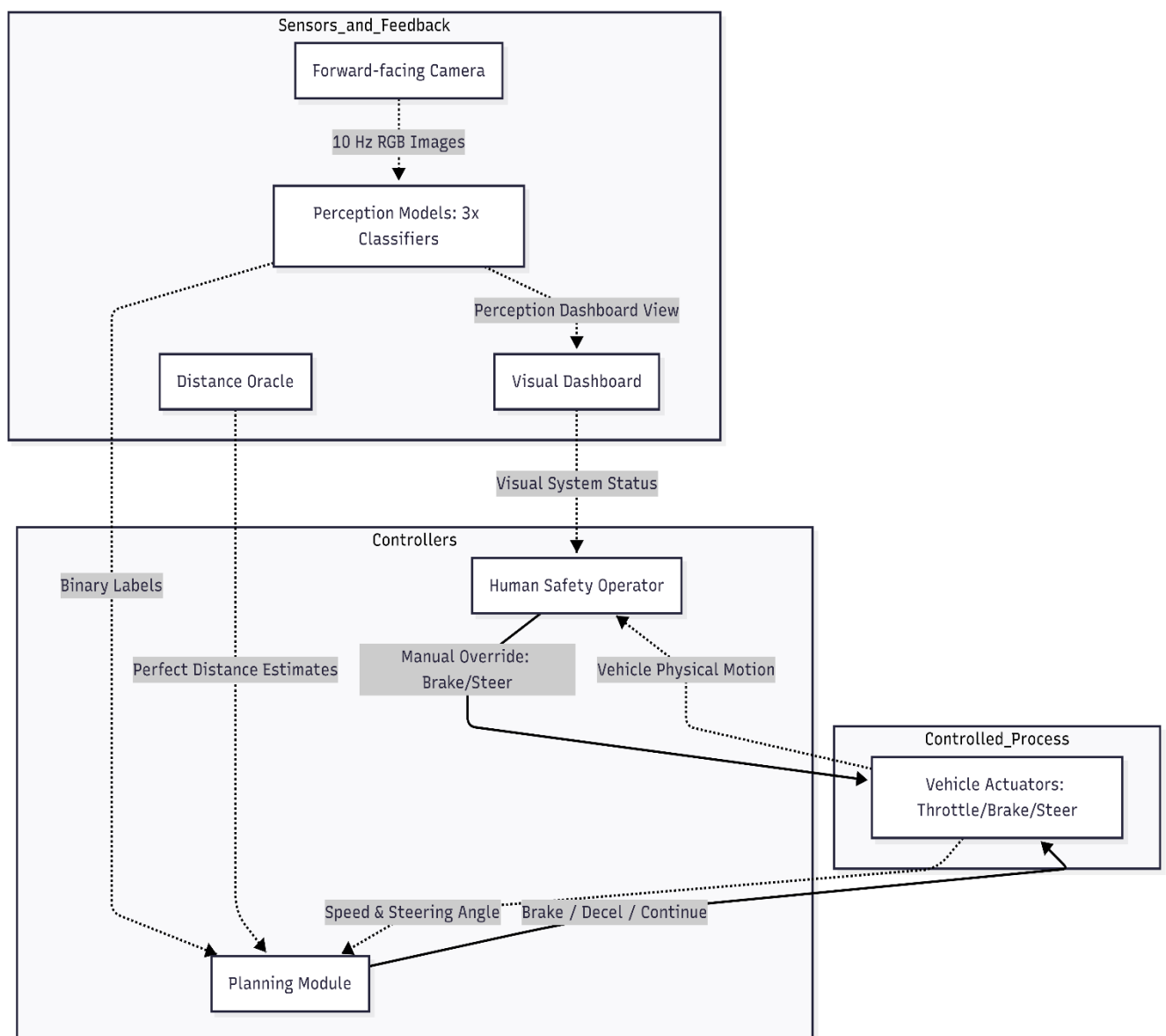
STPA Step 2: Model the Control Structure

Exercise 2.5: Control Structure Diagram

Draw the control structure for the CARLA system. Your diagram must include:

- All components listed in the system description (camera, perception models, distance oracle, planning module, actuators, human operator).
- **Control actions** (solid arrows): commands flowing from controllers to controlled processes (e.g., "brake command").
- **Feedback** (dashed arrows): information flowing back (e.g., "perception labels", "distance estimates", "vehicle speed").

Hint: There are at least two controllers in this system: the planning module and the human operator. The perception models and distance oracle should be considered *sensor/feedback*, not controllers themselves - they provide information to the controllers.



Hint: The traffic-light model only detects presence, not traffic light state. Consider what decisions the planner can (or cannot) safely make with this information.

"Instead of looking at isolated parts, this diagram maps out how 'loops' of control work—showing exactly what commands flow out (solid lines) and what feedback comes back (dashed lines) to keep the vehicle safe.

Control Structure Diagram (Ex 2.5), make sure to point out that the Planning Module is "blind" to raw camera images and only sees the labels, which is why your feedback arrows are dashed.

STPA Step 3: Identify Unsafe Control Actions and Derive Safety Constraints**Exercise 2.6: Unsafe Control Actions (UCAs)**

Using your control structure, identify **unsafe control actions** for at least **two different controllers**. For each UCA, also indicate which hazard(s) from Exercise 2.4 it could cause or lead to (reference them by ID, e.g., “H-1”, “H-2”).

For each controller, consider all four UCA types:

1. **Not provided** - a required action is missing.
2. **Provided unsafely** - the action is issued when it should not be.
3. **Wrong timing / order** - too early, too late, or out of sequence.
4. **Wrong duration** - applied too long or stopped too early.

Fill in the table below (add rows as needed):

ID	Controller	Control Action	UCA Type	Hazard	Unsafe Scenario
UCA-1	Planner	Continue Driving	Provided unsafely	H-1	Pedestrian detected in path but vehicle does not stop.
UCA-2	Planner	Emergency Brake	Not provided	H-1	Pedestrian within critical distance but brake not issued.
UCA-3	Planner	Emergency Brake	Wrong timing	H-3	Braking issued too late to stop at a detected intersection.
UCA-4	Operator	Manual Override	Not provided	H-2	Automation fails outside ODD; operator fails to take over.
UCA-5	Planner	Emergency Brake	Provided unsafely	H-4	Hard brake issued on highway with no obstacle, causing rear-end.
UCA-6	Planner	Request Decel	Not provided	H-3	Traffic light detected (presence) but no slowdown initiated.
UCA-7	Operator	Manual Override	Wrong timing	H-1	Operator reacts too late (> 1.5s) to a missed ML detection.
UCA-8	Planner	Continue Driving	Wrong timing	H-3	Continues through intersection after light presence was detected.

"Here, we analyze how a controller—either the AI or the human—can mess up by issuing a command too late, not issuing it at all, or issuing it when it shouldn't."

Exercise 2.7: Safety Constraints

For each UCA from Exercise 2.6, derive at least one **safety constraint** - a requirement that, if satisfied, prevents the UCA from occurring or limits its consequences.

Classify each constraint as either:

- **Model-level** - can be verified by testing or evaluating the corresponding perception model (this is what you will do in Sheets 4–9).
- **System-level** - requires a design measure beyond the model (e.g., redundancy, operator protocol, fallback behavior).

"We turn those 'mess-ups' into rules; if a control action is unsafe, the Safety Constraint is the mandatory requirement we set to ensure that action never happens."

Exercise 2.7: Safety Constraints

UCA	Safety Constraint	Level	Verification
UCA-1 Planner continues driving when a pedestrian is detected	SC-1: The pedestrian perception model shall reliably detect pedestrians in the vehicle's path, and the planner shall not continue driving when a pedestrian is detected within the critical distance.	Model-level	Test pedestrian detector using recall, precision, F1 and confusion matrix; verify planner response to pedestrian detections.
UCA-2 Emergency brake is not provided when pedestrian is within critical distance	SC-2: A pedestrian within the critical distance shall be detected with sufficiently high recall so that an emergency-brake command is triggered.	Model-level	Measure pedestrian-detector recall and false negatives; test critical-distance scenarios.
UCA-3 Emergency brake is issued too late at an intersection	SC-3: The traffic-light perception model shall detect traffic lights with sufficient reliability and sufficiently early for the planner to initiate braking before the vehicle reaches the stopping point.	Model-level	Test traffic-light detector recall/F1 and evaluate performance under relevant ODD conditions.
UCA-4 Operator fails to manually override automation outside the ODD	SC-4: When the vehicle operates outside the defined ODD, autonomous operation shall be disabled or the human operator shall be explicitly required to take control.	System-level	Verify ODD monitoring, fallback/mode-transition logic and operator takeover procedure.
UCA-5 Planner issues emergency brake when no obstacle exists	SC-5: Perception models shall minimize false-positive obstacle detections so that emergency braking is not triggered when no relevant obstacle is present.	Model-level	Evaluate precision, false-positive rate and confusion matrices for pedestrian/vehicle detectors.
UCA-6 Planner does not request deceleration after traffic-light detection	SC-6: The planner shall not treat the mere presence of a traffic light as evidence that stopping is required; a valid traffic-light state must be available before a stop decision is made.	System-level	Verify planner logic and requirements through scenario-based/system-level testing.
UCA-7 Operator	SC-7: The system shall provide sufficient	System-	Operator-procedure

reacts too late to missed ML detection	warning and monitoring mechanisms to enable the safety operator to intervene before a missed perception result can lead to a hazardous state.	level	validation, human-in-the-loop testing and intervention-time analysis.
UCA-8 Planner continues through an intersection after traffic-light presence is detected	SC-8: The planner shall not command continued driving through an approaching intersection unless the available traffic information is sufficient to establish that continuing is safe.	System-level	Verify planner decision logic using scenario-based tests covering traffic-light states and ambiguous detections.

STPA Step 4: Identify Causal Loss Scenarios Exercise

2.8: Causal Loss Scenarios

For each UCA from Exercise 2.6, identify **at least one causal scenario** - a specific mechanism or sequence of events that could cause the UCA to occur in practice.

For each scenario, describe:

1. **The causal factor:** What component failure, misunderstanding, or adverse condition triggers the UCA?
2. **Why it occurs:** What is the root cause or underlying reason?
3. **Connection to constraint:** Which safety constraint from Exercise 2.7 is designed to prevent or mitigate this scenario?

Hint: Consider the following causal mechanisms (from the lecture):

- **Incorrect feedback:** A sensor or perception model provides wrong information to the controller.
- **Flawed internal model:** The controller misinterprets correct information (e.g., low confidence treated as “no detection”).
- **Timing delay:** Information arrives too late for the controller to respond safely.
- **Human factor:** The human operator fails to intervene due to attention lapse, delayed reaction, or mode confusion.

Fill in the table below (add rows as needed):

Exercise 2.8: Causal Loss Scenarios

UCA	Causal Scenario	Root Cause	Related Constraint
UCA-1	A pedestrian is walking into the vehicle's path. Due to poor visual conditions or an unusual pedestrian appearance, the pedestrian classifier outputs " no pedestrian ". The planner therefore receives no pedestrian warning and continues driving.	Incorrect feedback: perception model produces a false negative.	SC-1
UCA-2	A pedestrian is within the critical braking distance, but the pedestrian classifier misses the pedestrian. Since the planner receives no pedestrian detection, the emergency-brake condition is never triggered.	Incorrect feedback: false-negative pedestrian detection.	SC-2
UCA-3	The vehicle approaches an intersection where braking is required. The traffic-light detector detects the light too late, so the planner receives the required information only when there is insufficient distance to stop safely.	Timing delay / incorrect feedback: traffic-light detection occurs too late.	SC-3
UCA-4	Weather or lighting suddenly changes from the valid ODD. The autonomous system continues operating even though perception reliability may be degraded. The operator notices the problem but does not take control in time.	Human factor + missing system fallback: operator must monitor the system continuously and may react too late.	SC-4
UCA-5	The camera image contains a visual pattern that the vehicle or pedestrian classifier incorrectly interprets as an obstacle. The planner receives a false obstacle	Incorrect feedback: perception model produces a false positive.	SC-5

	detection and commands maximum braking even though the road is clear.		
UCA-6	The traffic-light detector outputs " traffic light present. " Because the model only reports presence/absence rather than the traffic-light state, the planner cannot determine whether the light is red, yellow or green. A flawed planner rule may therefore fail to initiate appropriate deceleration.	Flawed internal model: planner treats traffic-light presence as sufficient information for a stopping decision.	SC-6
UCA-7	A perception model misses an obstacle. The operator sees the situation on the dashboard but reacts only after a significant delay because of prolonged monitoring or reduced attention.	Human factor: delayed or missed operator intervention.	SC-7
UCA-8	A traffic light is detected as present near an intersection, but the planner continues driving because the information does not contain the traffic-light state or because the planner incorrectly interprets the available information.	Flawed internal model / insufficient feedback: traffic-light presence is insufficient to determine whether continuing is safe.	SC-8

Connecting STPA to the Rest of the Course

Exercise 2.9: Mapping Constraints to Evidence

optional

Review the safety constraints you derived in Exercise 2.7 and the causal scenarios from Exercise 2.8. For each **model-level** constraint, identify which future exercise sheet will provide evidence (refer to the Course Roadmap).

For each **system-level** constraint, briefly describe what kind of evidence or argument you would need - even though you cannot run an experiment for it in this course.

"This final step connects our theoretical safety rules to the actual coding experiments we will perform later in the semester to prove our system is safe."

Model-Level Safety Constraint	Relevant Exercise Sheet(s)	Type of Evidence Provided
SC-1: Pedestrian detector must reliably detect pedestrians and prevent continued driving when a pedestrian is detected.	Sheet 4, Sheet 5	Sheet 4 provides pedestrian-model precision, recall, F1 and confusion matrix. Sheet 5 evaluates distribution shift and ODD coverage.
SC-2: Pedestrian within critical distance must be detected with sufficiently high recall.	Sheet 4, Sheet 5	Pedestrian recall and false-negative evidence from Sheet 4; ODD/data-distribution analysis from Sheet 5.
SC-3: Traffic-light detector must detect traffic lights sufficiently early/reliably for safe braking.	Sheet 4, Sheet 5	Traffic-light precision/recall/F1 and confusion matrix from Sheet 4; ODD coverage/distribution-shift evidence from Sheet 5.
SC-5: Perception models must minimize false-positive obstacle detections.	Sheet 4, Sheet 5	Precision and confusion matrices from Sheet 4; distribution-shift and ODD analysis from Sheet 5.

System-Level Safety Constraint	Relevant Evidence / Argument
SC-4: Automation must be disabled or operator takeover required outside the ODD.	Evidence should demonstrate correct ODD monitoring, mode-transition/fallback behavior and a defined operator takeover procedure. Sheet 9 can provide evidence that OOD conditions such as rain, fog and night can be detected, but the actual fallback mechanism is a system-level argument.
SC-6: Planner must not interpret mere traffic-light presence as a stopping requirement.	Requires a system-level argument and scenario-based testing of planner logic. The traffic-light classifier itself only predicts presence , so model performance alone cannot prove this constraint.
SC-7: Operator must be able to intervene before a missed ML detection becomes hazardous.	Requires human-in-the-loop evidence, intervention-time analysis and operator-procedure evidence. This cannot be established solely from perception-model metrics.
SC-8: Planner must not continue through an intersection unless available traffic information establishes that continuing is safe.	Requires system-level scenario testing and verification of planner decision logic, particularly because the traffic-light model provides presence rather than state information.